NAME

mbgrdtilemaker − Creates a set of overlapping square grids (tiles) from an original topography grid.

VERSION

Version 5.0

SYNOPSIS

mbgrdtilemaker **−−input=***inputGridFile* **−−output=***outputRoot* [**−−tile−dimension=***tileDimension* **−−tile−spacing=***tileSpacing* **−−verbose −−help**]

DESCRIPTION

Mbgrdtilemaker reads an input GMT/netCDF topography grid and breaks it up into a set of overlapping square grid tiles. Each tile has 50% overlap in both the x and y directions with its neighboring tiles, tiled outward from the southwest corner of the input grid. This tiling scheme is used to prepare topography for the MB-System TRN (terrain relative navigation) tools, which query the reference model tile by tile.

The size of each tile can be set either as a number of grid nodes per side (**−−tile−dimension**) or as a desired tile half−width in grid coordinate units, typically meters (**−−tile−spacing**); if **−−tile−spacing** is given it takes precedence and is used, together with the input grid's node spacing, to derive an equivalent tile dimension. If neither option is given, a default tile dimension of 2001 nodes per side is used. Given the tile dimension (and hence the tile width), **mbgrdtilemaker** computes the number of tiles needed in each direction to cover the full input grid at 50% overlap, and for each tile position extracts the corresponding subgrid of node values from the input grid (filling any nodes that fall outside the input grid with the input grid's no-data value).

Each tile is written as its own GMT grid file (named *outputRoot_nnnn*.grd inside the *outputRoot* output directory, which is created if it does not already exist), and a catalog file named "tiles.csv" is written into that same directory listing, for each tile, its future octree filename and the easting and northing of its center. The original input grid is also copied into the output directory as "source_grid.grd". Finally, **mbgrdtilemaker** invokes **mbgrd2octree** once for each generated tile grid to produce a matching TRN octree (".bo") file, so **mbgrd2octree** must be available in the execution search path.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help

This "help" flag cause the program to print out a description of its operation and then exit immediately.

−−input=*inputGridFile*

Sets the filename of the input topography grid to be tiled. This must be a GMT/netCDF grid readable by the **MBIO** grid I/O routines. This option (or the equivalent short option **−I**) must be specified.

−−output=*outputRoot*

Sets the root name used both for the output directory that is created to hold the tile grids, octree files, and "tiles.csv" catalog, and for the basenames of the individual tile files written into that directory. This option (or the equivalent short option **−O**) must be specified.

−−tile−dimension=*tileDimension***, −D***tileDimension*

Sets the number of grid nodes along each side of a (square) output tile. If neither this option nor **−−tile−spacing** is given, *tileDimension* defaults to 2001. If **−−tile−spacing** is also given, **−−tile−spacing** takes precedence and *tileDimension* is instead derived from it.

−−tile−spacing=*tileSpacing***, −S***tileSpacing*

Sets the desired tile half−width in the input grid's horizontal coordinate units (typically meters). When given, the tile dimension (number of nodes per side) is calculated from *tileSpacing* and the input grid's node spacing, overriding any value set with **−−tile−dimension**.

−−verbose

By default (**−−verbose** not given) **mbgrdtilemaker** prints only a modest summary: a line identifying the input grid and, at the end of the run, a line giving the number of tiles generated and the output location. Giving **−−verbose** (or **−V**) at least once raises the level of detail reported, causing **mbgrdtilemaker** to also print the full input grid parameters, the full output tileset parameters, and a line per tile and per **mbgrd2octree** invocation as the tiles are generated. Repeating **−−verbose** a second time additionally redirects all of this output to stderr instead of stdout and adds a further level of debug detail.

SEE ALSO

mbsystem(1), **mbgrd2octree**(1), **mbtrnpp**(1), **mbgrid**(1)

BUGS

Maybe. Maybe not.